

6 (a) From graph, resistance =  $1.85 \Omega$

(b) Effective resistance of the 4 components:

$$R_{\text{eff}} = \left[ \frac{1}{2.5 + 3.6} + \frac{1}{1.85 + 4.5} \right]^{-1} = 3.111 \Omega \quad [\text{C1 for working}]$$

Hence, terminal p.d.:

$$V_{\text{terminal}} = \left( \frac{3.111}{3.111 + 0.70} \right) 6.0 \quad [\text{M1}]$$

$$= 4.9 \text{ V} \quad [\text{A0}]$$

Alternative:

$$I = \frac{V}{R_{\text{total}}} = \frac{6.0}{3.111 + 0.70} = 1.5744 \text{ A}$$

$$V_{\text{terminal}} = E - Ir$$

$$= 6.0 - (1.5744)(0.70) \quad [\text{M1}]$$

$$= 4.9 \text{ V}$$

(c)

$$I = \frac{V_{\text{terminal}}}{1.85 + 4.5}$$

$$= \frac{4.9}{1.85 + 4.5} \quad [\text{M1}]$$

$$= 0.771 \text{ A} \quad [\text{A1}]$$

(d)  $V_{\text{thermistor}} = IR_T = (0.771)(1.85) = 1.43 \text{ V} \quad [\text{B1}]$

OR

$$V_{\text{thermistor}} = \frac{1.85}{1.85 + 4.5} (4.9) = 1.43 \text{ V} \quad [\text{B1}]$$

(d) For the galvanometer to read zero, the potentials at C and D must be the same (i.e. no p.d. between C and D).

For p.d. across CD to be zero,

$$V_{AC} = V_{\text{thermistor}}$$

$$V_{AB} = \left( \frac{2.5}{2.5 + 3.6} \right) (4.9) = 2.008 \text{ V} \quad [\text{C1 for working to show } V_{AC} = V_{\text{thermistor}}]$$

$$\frac{V_{AC}}{V_{AB}} = \frac{L_{AC}}{L_{AB}}$$

$$L_{AC} = \frac{V_{AC}}{V_{AB}} \times L_{AB} = \frac{1.43}{2.008} (1.0) = 0.71 \text{ m} \quad [\text{A1}]$$

OR

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For p.d. across CD to be zero,

$$V_{AC} = V_{\text{thermistor}}$$

$$\left( \frac{R_{AC}}{2.5 + 3.6} \right) (4.9) = 1.43 \quad [\text{C1 for working to show } V_{AC} = V_{\text{thermistor}}]$$

$$R_{AC} = 1.776 \, \Omega$$

$$L_{AC} = \frac{1.776}{2.5} (1.0) = 0.71 \, \text{m} \quad [\text{A1}]$$